

# Ch7\_Data Analysis Types

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## What is Data Analysis ?

<https://www.coursera.org/articles/what-is-data-analysis-with-examples>

1. **Identify** – business **question** you'd like to answer.

What problem is the company trying to solve?

What do you need to measure, and how will you measure it?

2. **Collect** – raw data sets

3. **Clean** – prepare it for analysis (duplicate/ anomalous/ inconsistencies/ standardizing data structure and format/ syntax errors)

4. **Analyze** – techniques and tools

→Find trends, correlations, outliers, and variations(变化).

→Using **data mining** to discover patterns within databases or **data visualization** software → graph

5. **Interpret** – 解释result 如何回答 business question

## 4 types of data analysis

- **Descriptive analytics**: “What happened?”
- **Diagnostic analytics**: “Why did it happen?”
- **Predictive analytics**: “What might happen next?”
- **Prescriptive analytics**: “What should we do next?”

1. **Descriptive analysis** 描述定量数据

### What in the past?

Tells us what **happened**. This type of analysis helps describe or summarize **quantitative data** by presenting statistics.

2. **Diagnostic analysis**

diagnostic analysis determines the “**why.**”

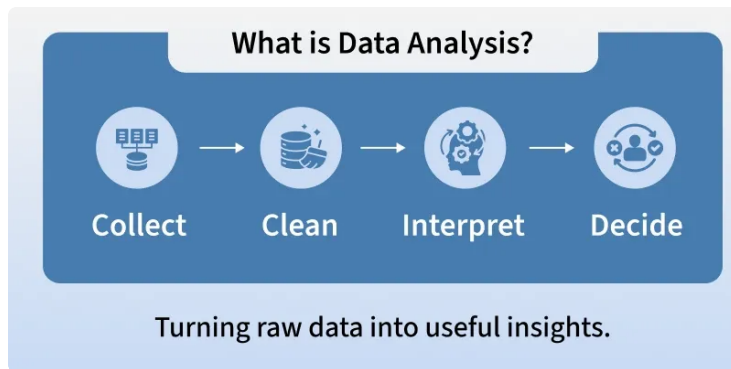
3. **Predictive analysis**

What **might happen in the future?**”

#### 4. Prescriptive analysis

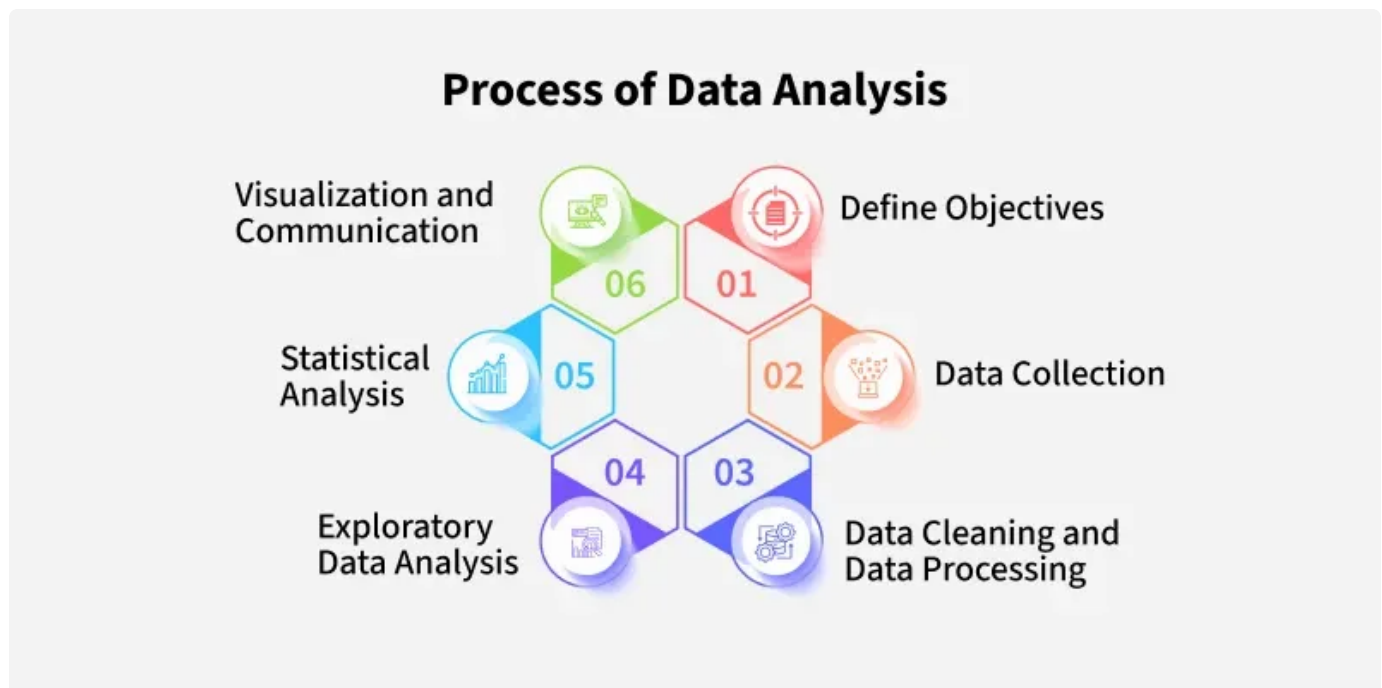
“What should we do about it?”

**Prescriptive analysis** takes all the insights gathered from the first three types of analysis and uses them to form recommendations for how a company should act.



Data analysis: RAW INFORMATION --> ACTIONABLE INSIGHTS

<https://www.geeksforgeeks.org/data-analysis/what-is-data-analysis/>



- **Exploratory Data Analysis (EDA):** Explore the data with visuals and summary statistics to spot patterns, trends and anomalies. This step helps refine our approach.
- **Statistical Analysis:** Apply statistical methods or models to test ideas, find relationships and make predictions that answer our original questions.

|        | Exploratory Data Analysis (EDA)        | Statistical Analysis                  |
|--------|----------------------------------------|---------------------------------------|
| 目的     | 发现数据中的模式、异常、分布特征                       | 验证假设、建立模型、推断关系                        |
| 阶段     | 分析初期，用于理解数据                            | 分析中后期，用于回答研究问题                        |
| 方法     | 可视化（图表）、描述性统计                          | 推断统计、回归、假设检验等                         |
| 是否假设驱动 | 非假设驱动（开放探索）                            | 假设驱动（目标明确）                            |
| 典型工具   | Histogram, Boxplot, Correlation Matrix | t-test, ANOVA, Regression, Chi-square |

## PPT CONTENT

### Three Levels of Analytics

- Descriptive Analytics
  - Summarizes historical data
  - Answers "What happened?"
- Predictive Analytics
  - Forecasts future trends
  - Answers "What could happen?"
- Prescriptive Analytics
  - Recommends optimal actions
  - Answers "What should we do?"

### 1. Descriptive Analytics

Descriptive Analytics（描述性分析）是数据分析的基础阶段，主要用于总结、可视化数据，识别趋势、异常值和变量关系。

#### 1. Summarizes and visualizes data

- Creates meaningful summaries of large datasets 汇总
- Uses statistical methods to describe key features 使用统计方法描述关键特征 → 包括中位数、平均数、标准差、分布形态、集中趋势等

#### 2. Explores main characteristics and patterns 探索主要特征与模式

- Identifies trends, outliers, and relationships
- Provides insights into data distribution 提供数据分布的洞察 → 例如是否呈正态分布、偏态分布，是否存在多峰结构。

### 3. Foundation for financial reporting and business analysis

#### 财务报告与商业分析的基础

- Supports regulatory **compliance** 支持合规性要求 → 确保数据报告满足审计、监管或行业标准。
- Informs initial business decision-making 为初步商业决策提供信息支持 → 帮助企业识别机会、优化资源配置、制定策略方向。

#### 1.1. Relationship: Scatter Plot (散点图)

A scatter plot is a graph that displays the relationship **between two numeric variables** using dots

- Useful for identifying correlations, trends, and outliers

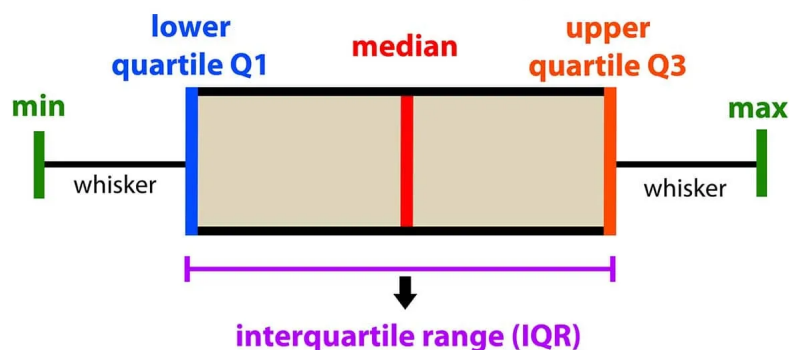
#### 1.2. Outliers: Box plot

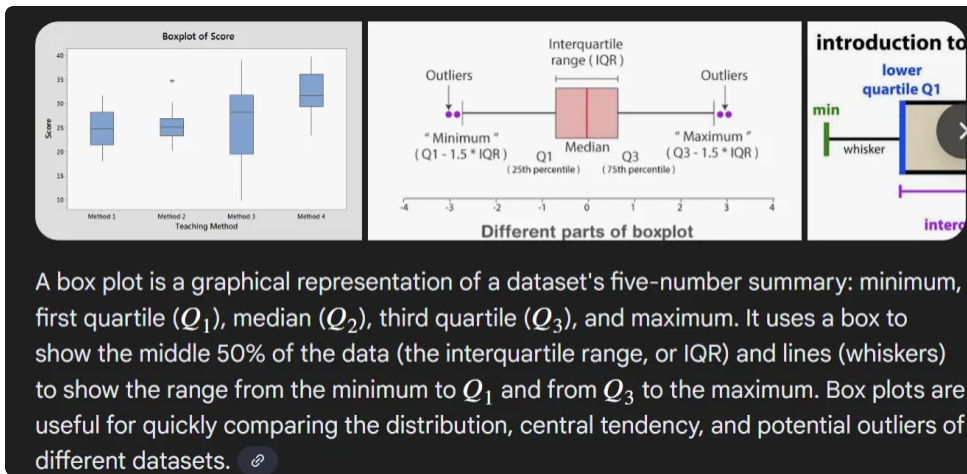
A box plot is a visual representation of a dataset's **distribution** that uses a **five-number summary** to show its median, quartiles, and range

Compare and construct two or more group

|      |        |                          |
|------|--------|--------------------------|
| 箱体较小 | IQR 较小 | 数据集中在中间 50% 区间，波动小，稳定性高  |
| 箱体较大 | IQR 较大 | 数据在中间 50% 区间分布广，波动大，稳定性低 |
| 箱体偏移 | 中位数不居中 | 数据偏态分布，可能左偏或右偏           |
| 箱体对称 | 中位数居中  | 数据分布较均匀，可能接近正态分布         |

### introduction to data analysis: Box Plot





### 1.3. Descriptive Analytics – Key Steps

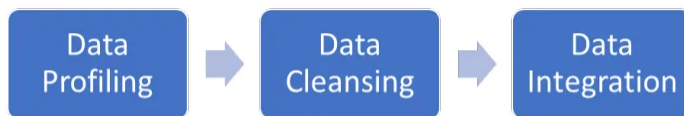


Figure 7.1 Preparing for Data Analysis

#### 1. Data Profiling (数据剖析) --> 提前理解data

Data profiling is the process of examining, analyzing, and creating summaries of data to understand its content, quality, and structure. It identifies data quality issues, such as inconsistencies, duplicates, and missing values, by using a combination of automated tools and analytical algorithms. The goal is to gain a comprehensive understanding of the data before it is used for analytics, business intelligence, or other data-driven projects.

Examines data structure and quality

Identifies data types, formats, and ranges

- Data structure: 数据的组织方式, 包括字段名称、数据类型、层级关系等

=> 决定数据是否可读、可处理。

#### 2. Data Cleansing

Corrects or removes errors and inconsistencies

Handles missing values and outliers

- eg. 用平均数来补充

### 3. Data integration

Combines data from multiple sources 整合来自多个来源的数据

Creates a unified view of the data 创建统一的数据视图

典型场景：

- 合并 CRM 与 ERP 数据
  - 整合线上与线下销售记录
  - 跨部门数据汇总（如财务 + 营销）
- 

## 1.4. Descriptive Analytics – Methods

- Outlier Detection
  - Identifies extreme or abnormal values
  - Uses techniques like z-scores and boxplots
- Normalization
  - Scales data to a specific range (e.g., 0 to 1)
  - Makes data more comparable across different scales
- Standardization
  - Transforms data to have mean 0 and standard deviation 1
  - Reduces the effect of outliers
- Imputation
  - Replaces missing values with estimates
  - Preserves statistical properties of the data

### 1. Outlier Detection（异常值检测）

- **Z-score**：衡量数据点与均值的偏离程度，通常超过  $\pm 3$  被视为异常。
- **Box Plot（箱线图）**：通过 IQR（四分位距）识别异常值。

### 2. Normalization（归一化）

将数据缩放到特定范围（如 0 到 1） -> 应用于机器学习模型前的数据准备

### 3. Standardization（标准化）

将数据转换为均值为 0、标准差为 1 的分布 -> 减少异常值的影响

### 4. Imputation（缺失值填补）

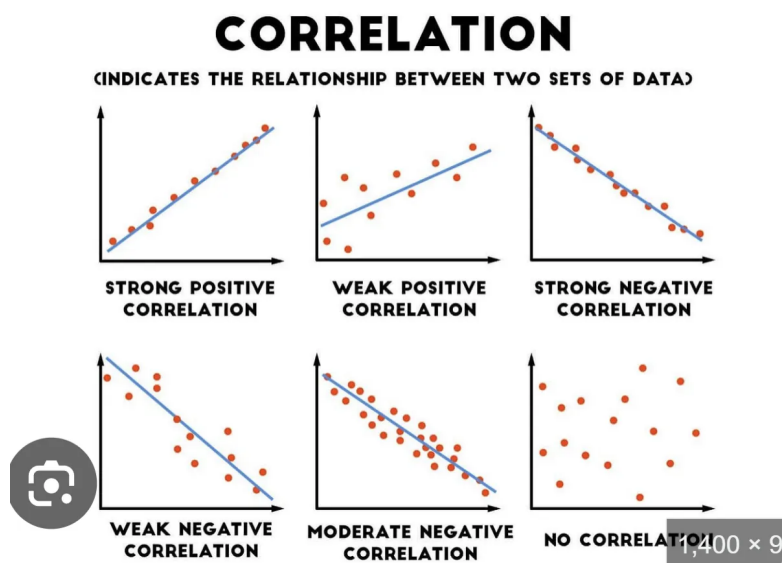
用估算值替代缺失数据

常见填补方式：

- 均值填补, 中位数填补, 插值法（如线性插值）, 模型预测填补（如 KNN、回归）

## 1.5. Descriptive Statistical Tools

- **Histograms**
  - Show frequency distribution of a variable
  - Visualize data shape and central tendency
- **Boxplots**
  - Display five-number summary of data
  - Identify outliers and data spread
- **Scatterplots**
  - Show relationship between two variables
  - Visualize correlation and patterns
- **Correlation Analysis**
  - Measures strength of relationship between variables
  - Quantifies linear dependencies in data



## 1.6. Feature Engineering

特征 (Feature) 是数据中用于描述对象属性、行为或状态的变量

| 特征名                     | 含义        | 示例值     |
|-------------------------|-----------|---------|
| 房屋面积 (Area)             | 房屋建筑面积    | 120 平方米 |
| 房间数 (Rooms)             | 房间数量      | 3       |
| 距地铁距离 (DistanceToMetro) | 离最近地铁站的距离 | 500 米   |
| 建造年份 (YearBuilt)        | 房屋建造年份    | 2010    |

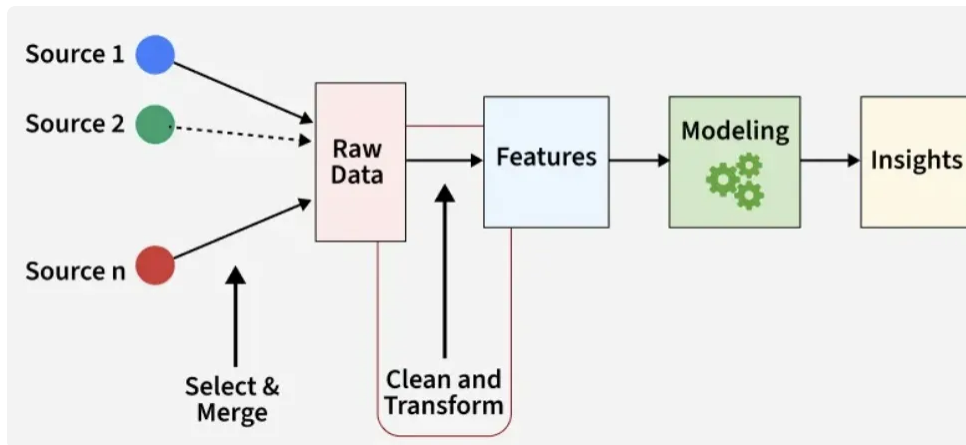


- Creating new variables from existing ones **what?**
  - Combines or transforms existing features
  - Extracts more meaningful information from raw data
- Enhances predictive power and interpretability **意义?**
  - Improves model performance
  - Makes complex relationships more apparent
- Examples: One-hot encoding, Binning
  - One-hot encoding: Converts categorical variables to binary
  - Binning: Groups continuous variables into discrete intervals

<https://www.geeksforgeeks.org/machine-learning/what-is-feature-engineering/>

RAW DATA → USEFUL FEATURE

The goal is to make the model better by providing relevant and easy-to-understand information.



### Common Techniques in Feature Engineering

1. **One-Hot Encoding:** One-Hot Encoding converts categorical variables into binary indicators, allowing them to be used by machine learning models.
2. **Binning:** Binning transforms continuous variables into discrete bins, making them categorical for easier analysis.



```

1 import pandas as pd
2
3 data = {'Age': [23, 45, 18, 34, 67, 50, 21]}
4 df = pd.DataFrame(data)
5
6 bins = [0, 20, 40, 60, 100]
7 labels = ['0-20', '21-40', '41-60', '61+']
8
9 df['Age_Group'] = pd.cut(df['Age'], bins=bins, labels=labels, right=False)
10
11 print(df)

```

### Output

|   | Age | Age_Group |
|---|-----|-----------|
| 0 | 23  | 21-40     |
| 1 | 45  | 41-60     |
| 2 | 18  | 0-20      |
| 3 | 34  | 21-40     |
| 4 | 67  | 61+       |
| 5 | 50  | 41-60     |
| 6 | 21  | 21-40     |

3. **Feature Splitting:** Divides a **single feature** into **multiple sub-features**, uncovering valuable insights and improving model performance.

```

1 import pandas as pd
2
3 data = {'Full_Address': [
4     '123 Elm St, Springfield, 12345', '456 Oak Rd, Shelbyville, 67890']}
5 df = pd.DataFrame(data)
6
7 df[['Street', 'City', 'Zipcode']] = df['Full_Address'].str.extract(
8     r'([0-9]+\s[\w\s]+),\s([\w\s]+),\s([\d+])')
9
10 print(df)

```

### OUTPUT

|   | Full_Address                   | Street     | City        | Zipcode  |
|---|--------------------------------|------------|-------------|----------|
| 0 | 123 Elm St, Springfield, 12345 | 123 Elm St | Springfield | 12345    |
| 1 | 456 Oak Rd, Shelbyville, 67890 | 456 Oak Rd | Shelbyville | 67890... |

|   |      |                     |                  |
|---|------|---------------------|------------------|
| ① | 数据剖析 | Data Profiling      | 识别字段类型、格式、范围、缺失率 |
| ② | 数据清洗 | Data Cleansing      | 修复错误、处理缺失值与异常值   |
| ③ | 数据整合 | Data Integration    | 合并多源数据，统一字段与粒度   |
| ④ | 特征工程 | Feature Engineering | 构造、选择、转换用于建模的特征  |
| ⑤ | 建模分析 | Modeling & Analysis | 应用统计模型或机器学习算法    |

### 特征工程的核心目标是：

将原始数据转换为对模型更有意义、更具预测力的特征集合。

#### 1. 特征选择 (Feature Selection)

删除无关或冗余特征

#### 2. 特征转换 (Feature Transformation)

- 标准化、归一化、对数变换、分箱 (binning)
- 示例：将“收入”取对数，减少极端值影响

#### 3. 特征构造 (Feature Construction / Feature Creation)

- 从现有变量中构造新特征
- 示例：
  - “年龄 × 活跃度” → 表示“高活跃老用户”
  - “月消费额 / 月访问次数” → 表示“平均客单价”

#### 4. 特征编码 (Feature Encoding)

- 类别变量转为数值 (如 One-Hot、Label Encoding)
- 示例：“性别” → 男=0，女=1

- 数据清洗是“修数据”，特征工程是“造特征”。
- 特征工程的核心是：让模型更容易学到有用的规律。

### 1.7. Reflection Questions:

## 1. How might descriptive analytics benefit your organization?

## 2. What challenges might you face in implementing descriptive analytics?

Analysis from different department.

1. Management people: Benefits for decision-making.(**Data-driven strategy and decision making** -> Culture)
2. Marketing department: **Deeper customer understanding**
3. Financial department: Monitor ( **Improved Financial performancee monitoring**)
4. Operational: **Enhance Operational efficiency**: supply chain(delivery time->Better communication; HR-> employee behavior)

Reasonable can be the answer

Challenge of implementing descriptive analytics:

=>About data -> ANY challenge of data?

1. **Data Quality and integrity** - accurate, inconsistent, error, missing  
=> Within the Orginzation
2. Coordination within different department => **Data integration and Technical infrastructure**( Integrate from different source and system)
3. **Culture resistance and change management**
4. **Skills Gap and Lack of expertise**

## 2. Predictive Analytics

- Builds models to forecast future outcomes
  - Uses historical data to identify patterns
  - Applies statistical and machine learning techniques
- Uses historical data to identify patterns
  - Learns from past trends and relationships
  - Extrapolates patterns to make predictions
- Supports proactive decision-making
  - Anticipates future events or behaviors
  - Allows organizations to prepare for potential scenarios

### 2.1. Key Concept

- Training and Test Datasets
  - Training data: Used to build the model
  - Test data: Used to evaluate model performance
- Model Fitting
  - Process of training a model on data
  - Involves adjusting model parameters to minimize error
- Bias-Variance Tradeoff
  - Bias: Error from oversimplified model
  - Variance: Error from model's sensitivity to fluctuations in training data

#### 1. Training and Test Datasets (训练集与测试集)

: 强调“训练是学习，测试是验证”

#### 2. Model Fitting (模型拟合) – 调整模型参数 -> 误差最小化

3.

|          |    |                          |
|----------|----|--------------------------|
| Bias     | 偏差 | 模型过于简单，无法捕捉数据规律，导致预测误差大  |
| Variance | 方差 | 模型过于复杂，对训练数据过度敏感，泛化能力差   |
| Tradeoff | 权衡 | 需在偏差与方差之间找到平衡点，避免过拟合或欠拟合 |

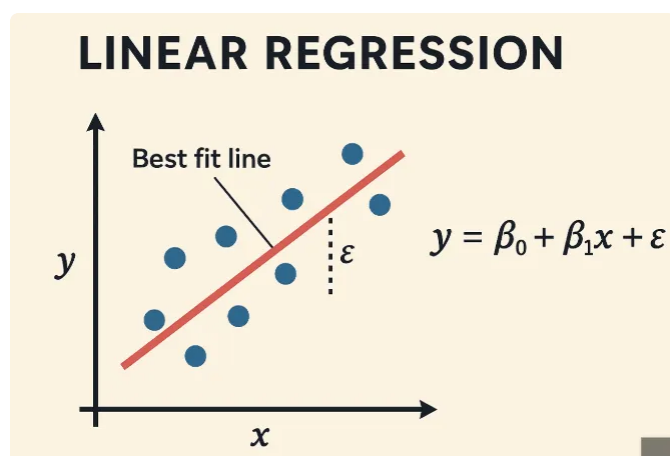
## 2.2. Methods

重点在于：需要了解各方法的特征 --> 判断特定情形如何使用。

### 1. Linear Regression => prediction 用于预测连续性结果

Linear relationship between variables 两个变量之间的关系可以用一次方程表示

Predicts continuous outcomes 是指可以取无限多个数值的结果变量



### 2. Logistic Regression => Classification 二分类问题

如，是否购买

Models probability of binary outcomes

信用违约预测、疾病诊断、用户流失分析

### 3. Decision Trees (决策树) => 可解释性强但易过拟合

Creates hierarchical decision rules

Easily interpretable but prone to overfitting -> 泛化能力差

### 4. Random Forests (随机森林) => 准确性高，适合复杂任务

Ensemble of decision trees

Improves accuracy and reduces overfitting

## 2.3. Advanced Techniques

### 1. K-means Clustering (K均值聚类)

Clustering  $\neq$  Classification, 是无监督学习, 用于发现数据中的自然分组

Groups similar data points into clusters 将相似的数据点分组为簇

- 根据距离将数据划分为 K 个组, 每组有一个中心点 (质心)
- Useful for customer segmentation

-> segmentation

classify different customer into different group

eg. bank, active/not -> search "customer segmentation in bank"

### 2. K-Nearest Neighbors (KNN) (K近邻算法)

Classifies based on similarity to nearby points 根据与邻近点的相似性进行分类

- Classifies based on similarity to nearby points
- Simple but effective for many problems

- 判断新用户是否会购买, 参考其行为最相似的历史用户

### 3. Support Vector Machines (SVM) (支持向量机)

Finds optimal hyperplane to separate classes 寻找最佳超平面以区分类别

- 最大化类别间的边界距离, 提高泛化能力
- High-dimensional data
- Finds optimal hyperplane to separate classes
- Effective for high-dimensional data

### 4. Neural Networks (神经网络)

✅ 神经网络可以用于:

- 连续型预测 (如房价、温度) → 回归任务
- 分类任务 (如图像识别、情感分析) → 二分类或多分类
- 生成任务 (如图像生成、文本生成) → 深度学习扩展

Models complex non-linear relationship

- 深度学习基础, 处理图像、语音、自然语言等
- 由多层神经元组成, 模拟人脑信息处理方式

## 2.4. Predictive Analytics — Evaluation

### 1. Cross-validation (交叉验证)

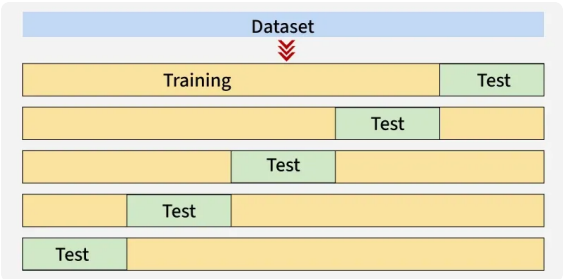
**Assesses model performance** on multiple data subsets

- Helps prevent overfitting

Check how well a machine learning model performs on unseen data while preventing overfitting. =>

Ensuring that all data points are used for both training and testing.

- Splitting the dataset into several parts.
- Training the model on some parts and testing it on the remaining part.
- Repeating this resampling process multiple times by choosing different parts of the dataset.
- Averaging the results from each validation step to get the final performance.



| Feature            | K-Fold Cross-Validation                                                         | Holdout Method                                                                           |
|--------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Data Split         | Dataset is divided into k folds and <u>each fold is used once</u> as test set   | Dataset is <u>split once</u> , typically into training and testing sets                  |
| Training & Testing | Model is trained and <u>tested k times</u> , each fold serving as test set once | Model is <u>trained once on training set and tested once on</u> test set                 |
| Bias & Variance    | Lower bias, more reliable performance estimate and variance depends on k        | <b>Higher bias</b> if the split is not representative and results can vary significantly |
| Execution Time     | Slower, especially for large datasets because model is trained k times          | <b>Faster</b> , only one training and testing cycle                                      |
| Best Use Case      | <u>Small to medium datasets where accuracy estimation is important</u>          | <u>Very large datasets or when quick evaluation is needed</u>                            |

**2. Grid Search (网格搜索)**

Systematically works through multiple model parameters 系统地遍历多个模型参数组合

- 寻找最佳参数组合以提升模型性能

**3. ROC Curves (ROC 曲线)**

Plots true positive rate against false positive rate 绘制真正率与假正率的关系图

- 作用：评估分类模型的性能
- 指标：AUC（曲线下面积）越大，模型越好

2.5. Reflection Question

1. How could predictive analytics improve forecasting in your industry?
2. What ethical considerations should be addressed when implementing predictive models?

|                       |                                                                                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Marketing 市场部         | Predictive analytics can <b>forecast customer behavior, ad effectiveness</b> 广告效果, <b>and market trends</b> . It helps identify which campaigns will succeed and which customer groups are most likely to buy. |
| Sales 销售部             | It can forecast sales volumes 销售额, customer demand, and churn risk 流失风险. By analyzing historical sales data, sales teams can set realistic targets and focus on high-value customers.                          |
| Operations 运营部        | Predictive analytics can forecast <b>inventory</b> needs 库存需求, production schedules 生产计划, and supply chain risks. This reduces stockouts or overstock 缺货或库存过剩 and optimizes resource use.                      |
| Human Resources 人力资源部 | It can predict employee turnover 流失率, recruitment needs, and training effectiveness. HR can plan ahead to stabilize workforce and reduce hiring risks.<br>人力资源部门能提前制定招聘和培训计划，稳定团队，降低风险                         |
| Finance 财务部           | Predictive analytics can forecast cash flow, ROI, and financial risks. This helps optimize budgets and improve capital efficiency.                                                                             |

|      |                 |             |
|------|-----------------|-------------|
| 维度   | 市场部 (Marketing) | 销售部 (Sales) |
| 关注点  | 市场和客户需求         | 客户购买和成交     |
| 工作性质 | 战略性、长期性         | 战术性、短期性     |
| 关系   | 为销售创造机会         | 把机会转化为收入    |

3. Prescriptive Analytics



### 3.1. Overview

- 是什么? **Recommends optimal actions** based on predictions
  - Integrates predictive insights with decision optimization
  - *Prescriptive analytics is the practice of analyzing data to identify patterns, which can be used to make predictions and **determine optimal courses of action***
  - *Prescriptive analytics adds a **recommendation layer** on top of predictive analytics, and differs from it in terms of focus, scope and approach.*
- 特征:
  - Uses **advanced techniques** to optimize decisions
  - Applies mathematical optimization and simulation 数学优化和模拟方法
  - Considers **multiple scenarios** and **constraints**
- 需求:
  - Requires robust data foundation and quality
    - Accurate and comprehensive data
    - **Various data sources and types**
      - Structured Data, Unstructured Data, Streaming Data, Time-Series Data

规范性分析是数据分析的最高层次，它不仅预测未来会发生什么，还进一步告诉你应该怎么做。它通过整合预测结果、优化算法和模拟技术，帮助企业在复杂约束下做出最优决策。

### 3.2. Methods

- **Simulation**
  - Models real-world systems and processes
  - Tests different scenarios and assumptions
- **Graph Analysis**
  - Analyzes relationships and connections in data
  - Optimizes network structures and flows
- **Complex Event Processing**
  - Analyzes real-time data streams
  - Detects and responds to patterns of events
- **Neural Networks**
  - Models complex non-linear relationships
  - Can be used for both prediction and optimization

1. Simulation 模拟现实系统和流程，测试不同情境与假设
  - 用于测试“如果.....会怎样”的情境，适合风险评估与资源调度。
2. Graph Analysis 分析数据中的关系与连接，优化网络结构与流动
  - 常用于物流、社交网络、交通等领域，优化路径与连接。

### 3. Complex Event Processing 复杂事件处理： 分析实时数据流，识别并响应事件模式

- **金融交易监控**：识别欺诈行为或异常交易模式
- **网络安全**：实时检测攻击行为或异常访问
- **电商推荐**：根据用户行为实时调整推荐内容
- **交通系统**：分析实时交通流，优化信号控制

### 4. Neural Networks 神经网络： 建模复杂的非线性关系，可用于预测与优化

- 处理复杂数据关系，既能预测，也能用于优化，例如在动态定价或个性化推荐中。

## 3.3. Prescriptive Analytics Project Lifecycle

### 1. Prescriptive Analytics Project Phases

1. **Requirements Analysis**
  - Gather and document business needs
  - Define project scope and objectives
2. **System Design**
  - Create technical architecture
  - Plan data flows and integrations
3. **Implementation and Testing**
  - Develop and integrate system components
  - Conduct thorough testing and validation
4. **Deployment and Maintenance**
  - Roll out the system to users
  - Provide ongoing support and updates

### 2. Managing the project Lifecycle

- **Initiation and Planning**
  - Define project charter and secure sponsorship
  - Create detailed project plans and timelines
- **Resource Management**
  - Allocate team members and skills
  - Manage budgets and equipment
- **Financial Governance**
  - Track expenses and ROI
  - Ensure compliance with financial policies
- **Compliance and Risk Management**
  - Address data privacy and security concerns
  - Mitigate project risks

### 3. Operationalizing Data Projects 数据项目的运营化

Data Project： 数据项目是指任何围绕数据展开的结构化工作 any structured initiative，包括数据的采集、处理、分析和应用，目的是解决问题、生成洞察或支持决策。它可以是构建数据仓库 building a data warehouse、开发机器学习模型 developing machine learning models、或设计可视化仪表盘 creating dashboards 等。

将数据项目的落地实施分为四个关键维度：

- Managing Massive Amounts of Data 管理海量数据

|                                             |              |
|---------------------------------------------|--------------|
| Implement <b>scalable</b> storage solutions | 实施可扩展的存储解决方案 |
| Optimize <b>data processing</b> pipelines   | 优化数据处理流程     |

随着数据量的爆炸式增长，企业必须采用可扩展的存储架构（如云存储、分布式文件系统 cloud storage and distributed file systems），并优化数据处理管道（如使用并行处理、批处理与流处理结合 parallel processing, batch processing, and stream processing），以确保数据能高效、安全地被采集、存储和分析。

- Addressing Data Silos

|                                                |             |
|------------------------------------------------|-------------|
| Integrate data from <b>various sources</b>     | 整合来自不同来源的数据 |
| Promote data <b>sharing across departments</b> | 促进部门间的数据共享  |

解决方法包括使用统一的数据平台、API接口、数据湖等技术

- Ensuring Data Quality

|                                                      |               |
|------------------------------------------------------|---------------|
| Implement <b>data governance</b> practices           | 实施数据治理机制      |
| Establish <b>data quality metrics and monitoring</b> | 建立数据质量指标与监控体系 |

- Coordinating Big Data/AI Initiatives

|                                                  |              |
|--------------------------------------------------|--------------|
| Align projects with overall <b>data strategy</b> | 将项目与整体数据战略对齐 |
| Foster <b>collaboration</b> between teams        | 促进团队间的协作     |

#### 4. Challenges in Data Projects

- Finding Specialized Skills
  - Recruit and retain data science talent
  - Provide ongoing training and development
- Integrating Diverse Data Sources
  - Overcome technical and format differences
  - Ensure data consistency and accuracy
- Balancing Speed and Accuracy
  - Deliver timely insights without compromising quality
  - Implement iterative development processes
- Ensuring Data Security and Privacy
  - Comply with regulations (e.g., GDPR)
  - Implement robust security measures

## Data Project Management

### 1.1. Overview

|                                                   |                |
|---------------------------------------------------|----------------|
| Emphasizes collaboration and iteration            | 强调协作与迭代        |
| Encourages cross-functional teamwork              | 鼓励跨职能团队合作      |
| Aligns data projects with strategic objectives    | 使数据项目与战略目标保持一致 |
| Allows for flexibility and continuous improvement | 允许灵活调整与持续改进    |
| Closely linked to business decisions              | 与业务决策紧密关联      |
| Balances technical and business perspectives      | 平衡技术视角与业务视角    |
| Emphasizes collaboration and iteration            | 强调协作与迭代        |
| Focuses on delivering actionable insights         | 关注输出可执行的洞察结果   |

数据项目管理不仅仅是传统项目管理的简单套用，它需要考虑数据的复杂性、实时性和跨部门协作。重点在于：

- **技术与业务并重**：既要懂数据架构，也要理解业务目标；
- **灵活迭代**：数据项目常常在探索中调整方向；
- **战略对齐**：数据分析必须服务于企业战略，而不是孤立存在；
- **可执行洞察**：最终目标是提供能指导行动的分析结果，而不是仅仅展示数据。

### 1.2. Agile Principles for Data Projects

|                                                       |                  |
|-------------------------------------------------------|------------------|
| Individuals and interactions over processes and tools | 重视个体与互动胜过流程与工具   |
| Prioritizes team communication and collaboration      | 优先考虑团队沟通与协作      |
| Values people over rigid methodologies                | 重视人而非僵化的方法论      |
| Working solutions over comprehensive documentation    | 优先交付可用解决方案而非冗长文档 |
| Focuses on delivering functional data products        | 关注交付可用的数据产品      |
| Emphasizes practical results over extensive paperwork | 强调实际成果而非繁复文书     |
| Customer collaboration over contract negotiation      | 客户协作优先于合同谈判      |
| Involves stakeholders throughout the project          | 项目全过程中都要让利益相关者参与 |
| Adapts to changing requirements and needs             | 适应不断变化的需求与目标     |
| Responding to change over following a plan            | 响应变化优先于严格遵循计划    |
| Embraces flexibility in project execution             | 在项目执行中拥抱灵活性      |
| Allows for pivots based on new insights or challenges | 根据新洞察或挑战进行方向调整   |

敏捷原则在数据项目中尤其重要，因为数据分析往往充满不确定性和探索性。关键点包括：

- **以人为本**：数据科学家、工程师、业务人员的协作是成功关键；
- **快速迭代**：数据产品应快速上线、持续优化；
- **灵活应变**：面对新数据、新需求，要能快速调整方向；
- **客户参与**：让业务方持续参与，确保分析结果有用、可落地；
- **减少文档负担**：重视实际交付而非形式主义。

### 1.3. SCRUM Framework for Data Projects

SCRUM 是一种敏捷项目管理方法，强调**短周期迭代**、**团队协作**和**持续反馈**。在数据项目中，它帮助团队快速交付可用的数据产品，并根据反馈不断优化。

|                |                                                                              |                    |
|----------------|------------------------------------------------------------------------------|--------------------|
| <b>Sprints</b> | Short, time-boxed periods for completing work<br>Usually 1–4 weeks in length | 短周期冲刺阶段，用于集中完成特定工作 |
|----------------|------------------------------------------------------------------------------|--------------------|

|                              |                                                                                       |                              |
|------------------------------|---------------------------------------------------------------------------------------|------------------------------|
| <b>Daily Stand-ups</b>       | Brief daily team meetings<br>Share progress, plans, and obstacles                     | 每日站会，快速同步进度、计划和遇到的障碍         |
| <b>Sprint Reviews</b>        | Demonstrate completed work to stakeholders<br>Gather feedback and adjust priorities   | 向利益相关者展示成果，收集反馈并调整优先级        |
| <b>Sprint Retrospectives</b> | Reflect on team processes and performance<br>Identify improvements for future sprints | 冲刺回顾，反思团队流程与表现，提出改进建议用于下一轮冲刺 |

SCRUM 框架下的数据项目通常涉及**三个核心角色**，每个角色承担不同职责，确保项目顺利推进。

|                          |                                                                                                      |                                    |
|--------------------------|------------------------------------------------------------------------------------------------------|------------------------------------|
| Product Owner<br>产品负责人   | Represents stakeholders and defines project vision<br>Prioritizes backlog and ensures value delivery | 代表利益相关者，定义项目愿景；负责任务优先级排序，确保交付有价值成果 |
| Scrum Master<br>敏捷教练     | Facilitates Scrum process and removes obstacles<br>Coaches team on Agile practices                   | 促进 SCRUM 流程，清除障碍；指导团队实践敏捷原则        |
| Development Team<br>开发团队 | Cross-functional group that executes the work<br>Includes data scientists, engineers, analysts, etc. | 跨职能执行团队，包含数据科学家、工程师、分析师等，负责实际工作执行  |